

Stop coding!

Advance in seniority

Wolfgang Gassler, Tech Talks South Tyrol, 2023-11-22



Wolfgang Gassler

Born in Innsbruck, Austria

Computer Science, Ph.D

- 25 years in industry (Perl, PHP3) 
- 10 years Academic Researcher, Lecturer, PM 
- 2016 Engineering Manager & Coach 
- Since 2020 Tech Coach & Consultant



Netflix

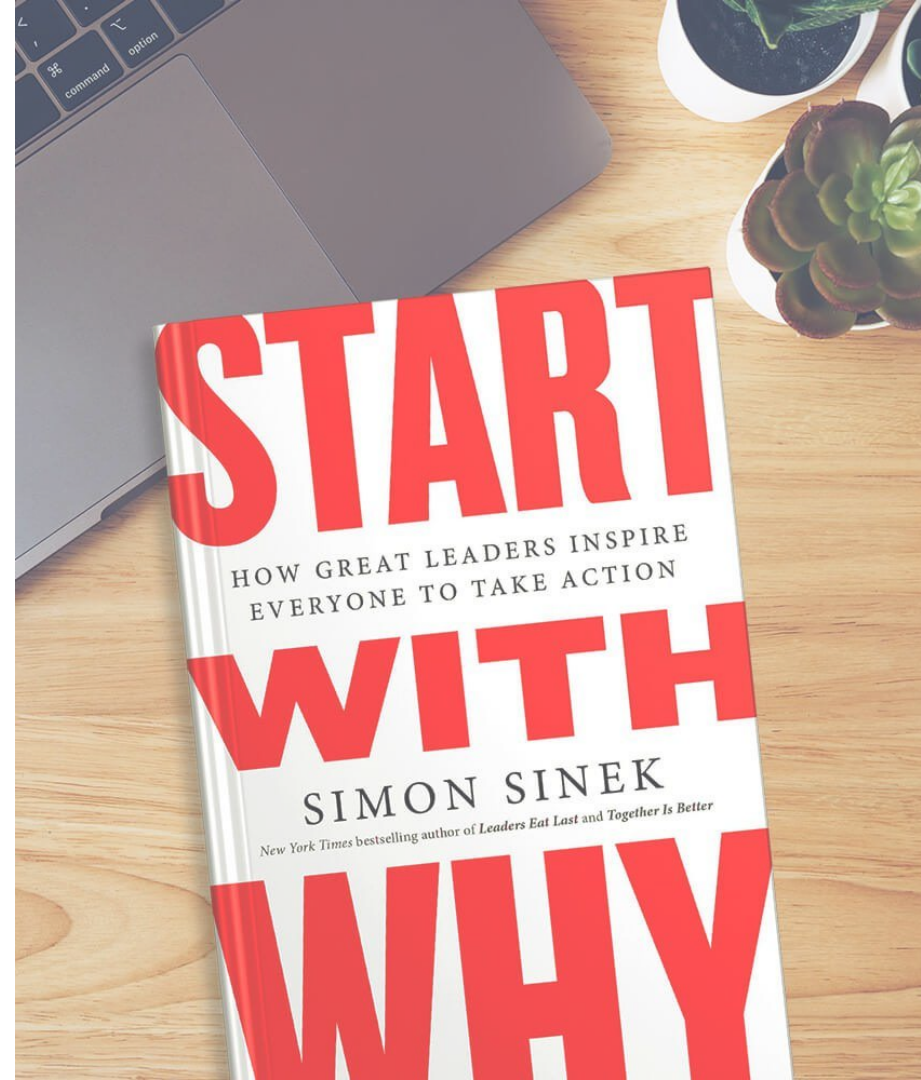
We model ourselves on being a professional sports team, not a family. A family is about unconditional love. A dream team is about pushing yourself to be the best possible teammate, caring intensely about your team, and knowing that you may not be on the team forever. **Dream teams are about performance, not seniority or tenure.** It is up to the manager to ensure that every player is amazing at their position, plays effectively with others and is given new opportunities to develop.

(Source: <https://jobs.netflix.com/culture>)

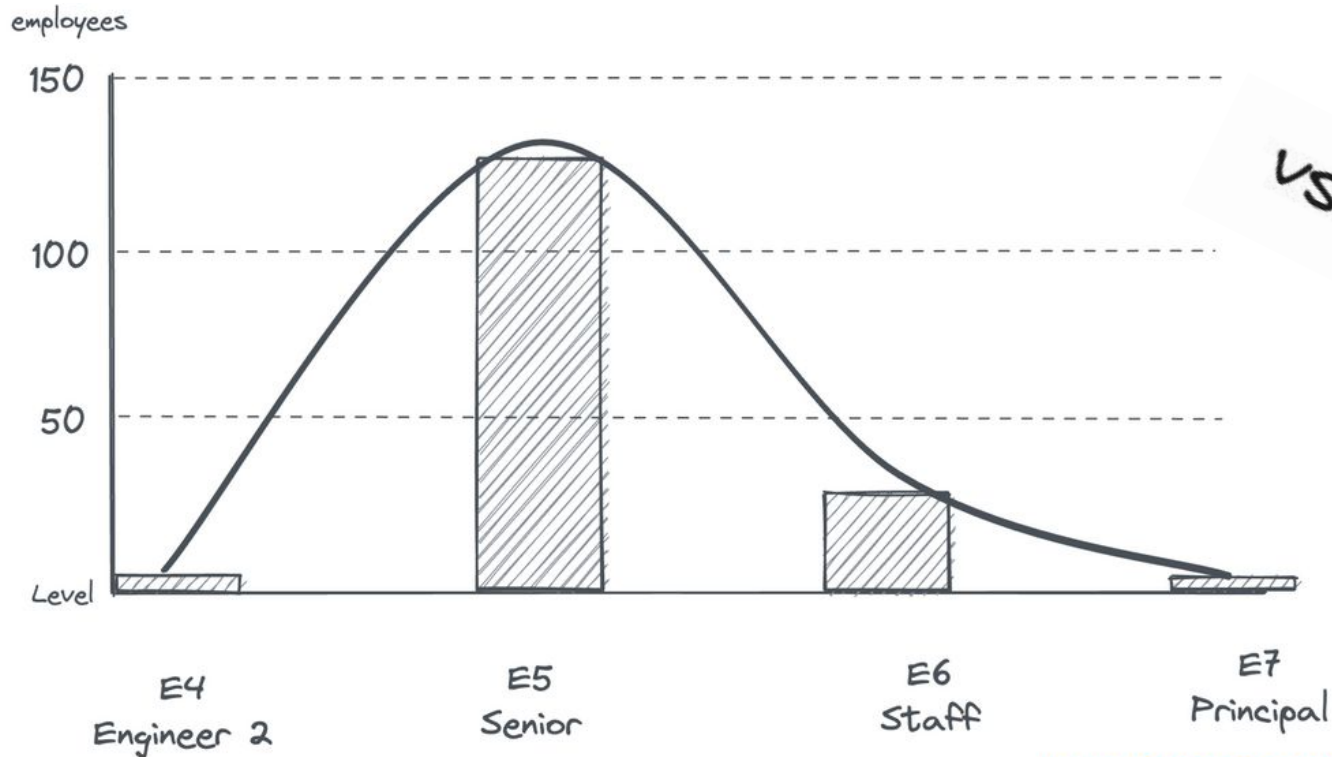


Why Levels?

- Cost
- Inability in hiring
- Expectations
- Transparency
- Recognition/appreciation
- Representation of underrepresented groups



Career path for individual contributors



Titles



Money doesn't matter
- Rich people



Looks doesn't matter
- Attractive people



"Marks doesn't matter"
- Toppers



Career Paths

How to grow as a individual contributor

Dropbox Engineering Career Framework

Overview

What is Impact?

▼ Software Engineer (SWE)

IC1 Software Engineer

IC2 Software Engineer

IC3 Software Engineer

IC4 Software Engineer

IC5 Staff Software Engineer

IC6 Principal Software Engineer

IC7 Senior Principal Software Engin...

SWE: Appendix

► Quality Engineer (QE)

► Software Development Engineer in Test...

► Reliability Engineer (SRE)

► Machine Learning Engineer (MLE)

► Data Engineer (DE)

► Security Engineer (SE)

Dropbox Engineering Career Framework

What's a Career Framework?

The Engineering Career Framework is your source for how to achieve impact for your role and team and how to grow in your engineering career. For managers, it can help you set expectations with your teams and hold them accountable for their work.

What the Career Framework is not

This framework is not a promotion checklist for your role; rather, it's designed to help you figure out what your impact could look like at the next level.

This framework is not an exhaustive list of examples and behaviors; each responsibility includes three to four key behaviors that serve as guide for how to think about your work. Consequently, you'll need to meet with your manager to define your impact goals and align on the expectations for your role.

What's in a Career Framework?

This framework is broken down into two components:

- **Level Expectations** define the scope, collaborative reach, and levers for impact at every level; these expectations are the *what* that determines the difference between an IC3 and IC4, for example
- **Core and Craft Responsibilities** define the key behaviors specific to your role and team; these behaviors help you identify *how* you work to deliver impact based on your level expectations

How to navigate this framework

Dropbox measures the success of its engineers largely on business impact. Anchor your work first and foremost on creating long-term impact. Since impact can be a bit vague, read [What is Impact?](#)

Next, ground yourself in the expectations for your level and team. For each level, you'll find a one-line summary description and the role's scope, collaborative reach, and levers for impact.

Review the expected behaviors for that level across the Results, Direction, Talent, Culture pillars from the Core Responsibilities. Read your Craft expectations, which are the per discipline technical capabilities you need to master at that level. Finally, meet with your manager to set your goals for the quarter.



IC1 Software Engineer

I deliver lots of high quality production-ready code with direction from the team

Scope

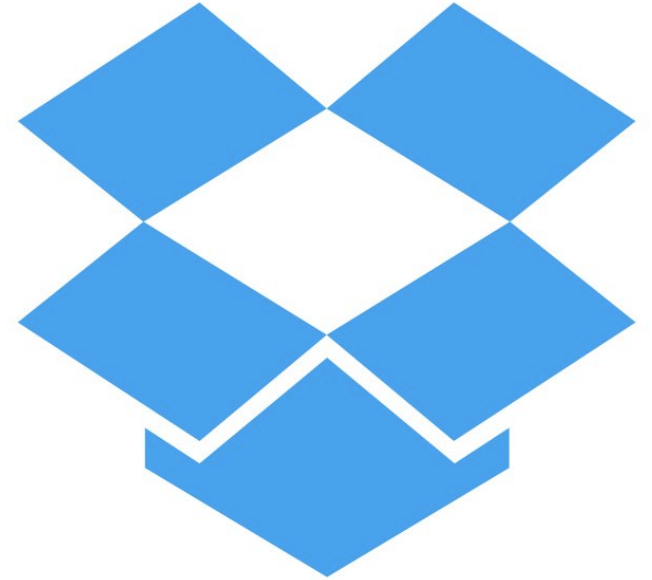
I execute on defined tasks and contribute to solving problems with defined solutions.

Reach

I work within the scope of my team with specific guidance from my manager/TL

Impact

Craft - I primarily focus on improving my craft as an engineer



Dropbox

E2 - Engineer

Technical skills: Consistently writes functions that are easily testable, easily understood by other developers, and accounts for edge cases and errors. Uses docstrings effectively.

Delivery: Understands and acts according to task prioritisation. Notes dependencies.

Feedback, Communication, Collaboration: Delivers praise and constructive feedback to their team, teammates, and manager in a useful manner.

Leadership: Sometimes thinks about team practices and processes and discusses improvements with team.

Strategic Impact: Has a basic understanding of their team's domain, and how it contributes to overall business strategy.

(just picked one per category as example)



Senior?



Stop coding!

Zooming out to bird's view

Big picture





Execution



Execution

Energy

Skills growth

Social capital

Credibility

Quality of life

Leveling up



New skills: Leading by influence

- You need all the tech skills and...
- Communication
 - Explaining
 - Listening
 - Delivering Messages (intern/extern)
 - Marketing
 - Writing
- Get used to horrible feedback cycles
- No direct impact (multiplier effect)
- Only do what others can't do



Stop coding!

Big Picture, Execution, and Leveling-Up

Stop Coding! Resources

EngineeringKiosk.dev Podcast (German) by Andy Grunwald & me

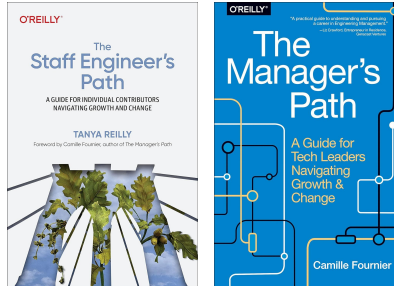
- [#51 Was ist das Staff \(Engineer\) Level?](#)
- [#47 Wer Visionen hat, soll zum Arzt!?](#)
- [#40 Wie wird man und Frau zum Senior Dev?](#)
- [#95 Effiziente Knowledge Sharing Formate: Wissen teilen und begeistern](#)

Articles

- [How to Exhibit Leadership as an Individual Contributor by Tom Bartel](#)

Books

- The Manager's Path by Camille Fournier
- The Staff Engineer 's Path by Tanya Reilly
- Staff Engineer: Leadership beyond the management track by Will Larson



Advance in seniority

Big Picture

Execution

Leveling-Up

Credits

- Pics by [David Brooke Martin](#), [Maria Lysenko](#), [Paul Skorupskas](#), [Javier Allegue Barros](#), [Marek Studzinski](#), [Apostolos Vamvouras](#), [Dylan Collette](#) on [Unsplash](#)
- Three pillars Big Pic/Execution/Leveling-up by Tanya Reilly
- [Dropbox Career Framework](#)
- [CircleCI Career Framework](#)

Additional Material

IC5 Staff Software Engineer

I set the multi-year, multi-team technical strategy and deliver it through direct implementation or broad technical leadership

Scope: *I deliver multi-year, multi-team product or platform goals. I exhibit a very high standard of technical judgement, innovation and execution to tackle open-ended problems that require difficult prioritization, defining both the what and how of things to be done*

Reach: *I am increasingly influencing the roadmaps of other Dropbox teams to achieve business impacting goals. I exercise judgement that favors the priorities of the wider engineering org rather than favoring locally optimal outcomes*

Impact: *Domain Expertise - I demonstrate a high level of depth in a particular platform or product category that brings unique business value*

Technical Strategy - I excel at defining the vision for and delivering large business-impacting projects with multiple constraints

Product Expertise - I increasingly influence or make product decisions/scope and determine the right technical tradeoffs to deliver customer value quickly.

Mentorship - I serve as a role model for other Dropbox engineers





E5 - Senior Staff Engineer

Delivery: Reviews cross-team work critically and ensures it's appropriately broken down and prioritised, and well understood by all involved teams.

Feedback, Communication, Collaboration: Consistently works across teams to help them resolve blockers, and complete work tasks. Ensures that credit is shared and given where due.

Leadership: Mentors across teams in an open, respectful, flexible, empathetic manner. Fosters a culture of mentoring across teams by seeking out mentoring opportunities for themselves and others, and supports others in their growth as mentors.

Strategic Impact: Has a thorough understanding of several team's domains, and how they contribute to overall business strategy.

(just picked one per category as example)



Tech Lead



Engineering
Manager

Engineering Manager

- Team Leadership
- Project Management
- Resource Allocation
- Performance Management
- Cross-Functional Collaboration
- Budgeting and Cost Management
- Risk Management
- Process Improvement
- Conflict Resolution
- Talent Acquisition

Tech Lead

- Technical Expertise
- Code Review
- Architecture and Design
- Task Distribution
- Mentorship
- Problem Solving
- Technical Documentation
- Collaboration
- Code Standards
- Innovation



ChatGPT

Engineering Manager

Team Leadership

Project Management

Resource Allocation

Performance Management

Cross-Functional Collaboration

Budgeting and Cost Management

Risk Management

Process Improvement

Conflict Resolution

Talent Acquisition

Tech Lead

Technical Expertise

Code Review

Architecture and Design

Task Distribution

Mentorship

Problem Solving

Technical Documentation

Collaboration

Code Standards

Innovation



ChatGPT